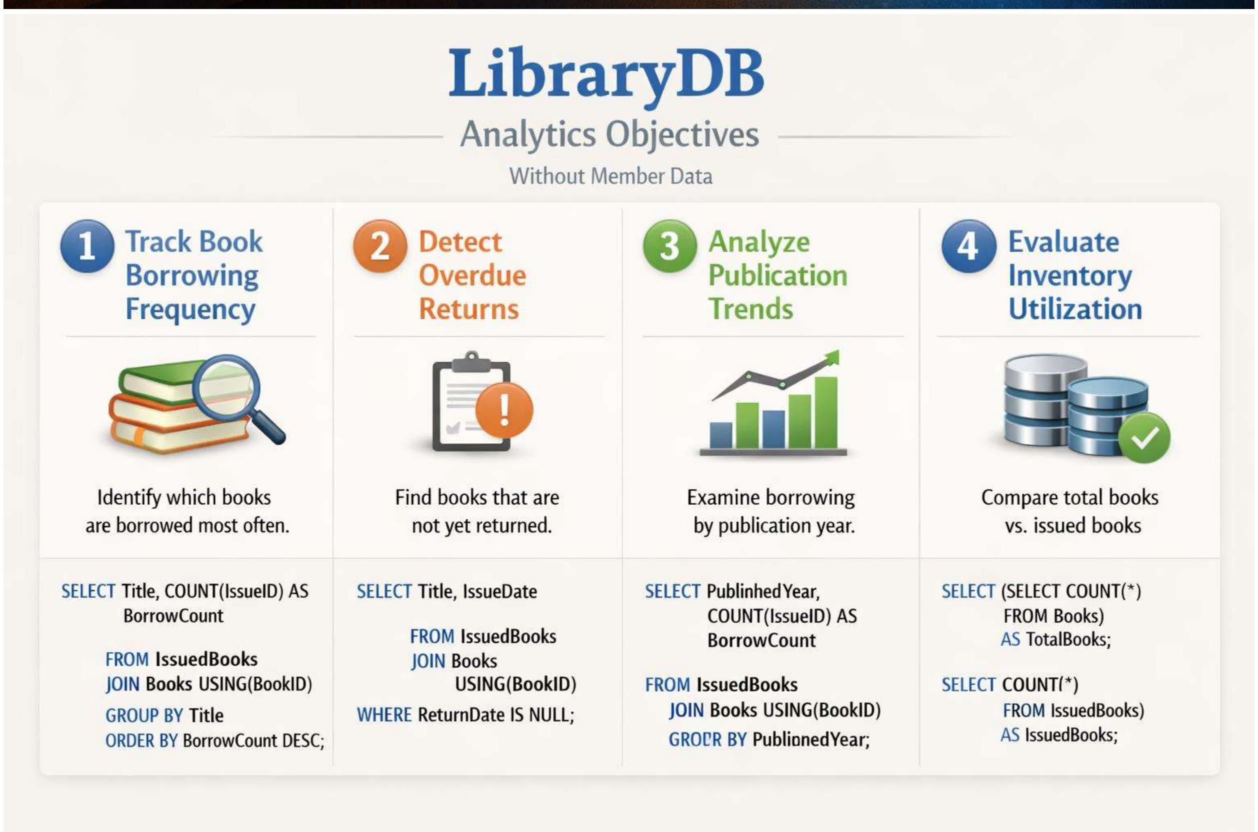

Library
Management
SQL Project
SQL
Book_ID
Title
Author
Status


LibraryDB
Analytics Objectives
Without Member Data
1 Track Book Borrowing Frequency
Identify which books are borrowed most often.
SELECT Title, COUNT(IssueID) AS BorrowCount
FROM IssuedBooks
JOIN Books USING(BookID)
GROUP BY Title
ORDER BY BorrowCount DESC;
2 Detect Overdue Returns
Find books that are not yet returned.
SELECT Title, IssueDate
FROM IssuedBooks
JOIN Books USING(BookID)
WHERE ReturnDate IS NULL;
3 Analyze Publication Trends
Examine borrowing by publication year.
SELECT PublinhedYear, COUNT(IssueID) AS BorrowCount
FROM IssuedBooks
JOIN Books USING(BookID)
GROCR BY PublinnedYear;
4 Evaluate Inventory Utilization
Compare total books vs. issued books
SELECT (SELECT COUNT(*) FROM Books) AS TotalBooks;
SELECT COUNT(*) FROM IssuedBooks) AS IssuedBooks;

🎯 Objective

The **LibraryDB** project aims to design a structured SQL database and an interactive dashboard for analyzing library operations. It focuses on tracking book borrowing patterns, overdue returns, and publication trends to support data-driven decision-making in library management.

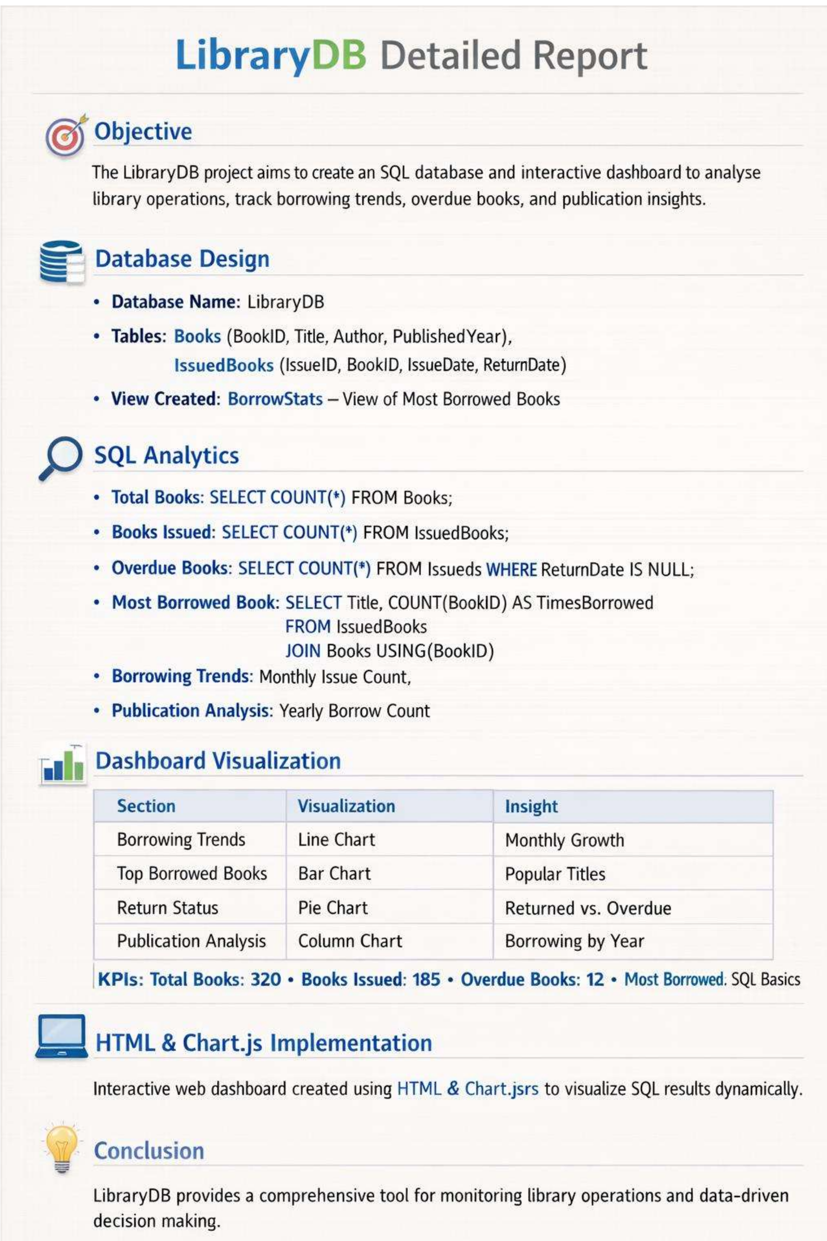
🧩 Database Design

- **Database Name:** `LibraryDB`
- **Tables:**
  - Books — stores book details (Title, Author, PublishedYear).
  - `IssuedBooks` — records issued and returned books with dates.
- **Relationships:** One-to-many link between Books and `IssuedBooks` via BookID.
- **View Created:** `BorrowStats` — summarizes how many times each book has been

💻 SQL Analytics

Key queries were developed to extract insights:

- **Total Books:** `SELECT COUNT(*) FROM Books;`
- **Books Issued:** `SELECT COUNT(*) FROM IssuedBooks;`
- **Overdue Books:** `SELECT COUNT(*) FROM IssuedBooks WHERE ReturnDate IS NULL;`
- **Most Borrowed Book:** Aggregates borrow counts per title.
- **Borrowing Trends:** Monthly issue frequency.
- **Publication Analysis:** Borrowing by publication year.



LibraryDB Detailed Report

Objective

The LibraryDB project aims to create an SQL database and interactive dashboard to analyse library operations, track borrowing trends, overdue books, and publication insights.

Database Design

- **Database Name:** LibraryDB
- **Tables:** Books (BookID, Title, Author, PublishedYear),
  IssuedBooks (IssueID, BookID, IssueDate, ReturnDate)
- **View Created:** BorrowStats – View of Most Borrowed Books

SQL Analytics

- **Total Books:** SELECT COUNT(*) FROM Books;
- **Books Issued:** SELECT COUNT(*) FROM IssuedBooks;
- **Overdue Books:** SELECT COUNT(*) FROM Issueds WHERE ReturnDate IS NULL;
- **Most Borrowed Book:** SELECT Title, COUNT(BookID) AS TimesBorrowed
  FROM IssuedBooks
  JOIN Books USING(BookID)
- **Borrowing Trends:** Monthly Issue Count,
- **Publication Analysis:** Yearly Borrow Count

Dashboard Visualization

Section	Visualization	Insight
Borrowing Trends	Line Chart	Monthly Growth
Top Borrowed Books	Bar Chart	Popular Titles
Return Status	Pie Chart	Returned vs. Overdue
Publication Analysis	Column Chart	Borrowing by Year

KPIs: Total Books: 320 • Books Issued: 185 • Overdue Books: 12 • Most Borrowed. SQL Basics

HTML & Chart.js Implementation

Interactive web dashboard created using HTML & Chart.jsrs to visualize SQL results dynamically.

Conclusion

LibraryDB provides a comprehensive tool for monitoring library operations and data-driven decision making.